

From eq. (2.17), $a_k = 0$ for k even and nonzero. Also $\sin(\pi k / 2)$ alternates between +1 and -1 for successive odd values of k . Therefore,

$$a_1 = a_{-1} = \frac{1}{\pi}, \quad a_3 = a_{-3} = -\frac{1}{3\pi}, \quad a_5 = a_{-5} = \frac{1}{5\pi} \quad \dots$$

The plot of Fourier Series coefficients a_k as a function of k or ω is known as the **spectrum** of the signal $x(t)$.

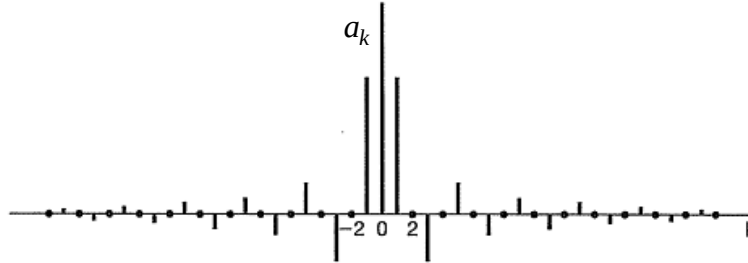


Figure 2.21. Plots of the Fourier series coefficients a_k for the periodic square wave with $T = 4T_1$. The coefficients are regularly spaced samples of the envelope $(2 \sin \omega T_1) / \omega T$ at $\omega = k\omega_0$.

Note that the Fourier Series coefficients is complex in general. Therefore, the spectrum of a general periodic signal consists of two plots: The plots of the spectral amplitude $|a_k|$ and the spectral phase $\angle a_k$.

2.7 Fourier series approximation to periodic functions

So far, we have only claimed that we can express any periodic function as a summation of sinusoids with frequencies that are integer multiples of $2\pi / T$. However, is that really true? Do we actually get the original function back by summing all these sinusoids?

In practice, we cannot sum an infinite number of sinusoids, so summing a finite number of sinusoids will always be an approximation to the original function $g(t)$. In other words,

$$x_N(t) \approx \sum_{n=-N}^N a_n e^{jn\frac{2\pi}{T}t}$$

where N is finite. The following figures show how the original function $x(t)$ compares with its Fourier representation as a function of N .

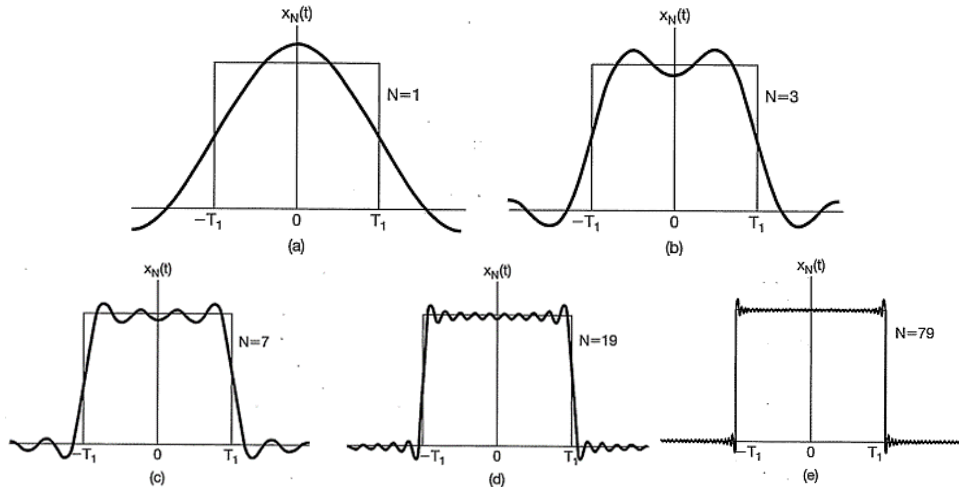


Figure 2.22. Convergence of the Fourier series representation of a square wave. Here, we have depicted the finite series approximation

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t} \text{ for several values of } N.$$

Example 2.10

Find the complex Fourier coefficients c_n of the following periodic function $f(t)$.

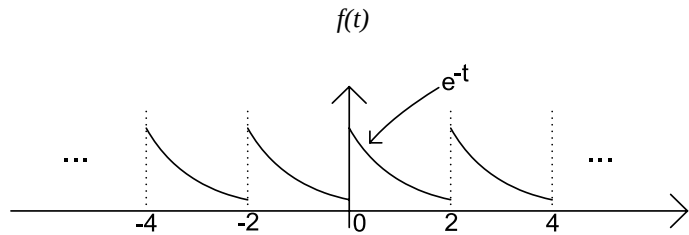


Figure 2.23 Periodic function of $f(t) = e^{-t}$ of period 2

Solution: We know from definition that,

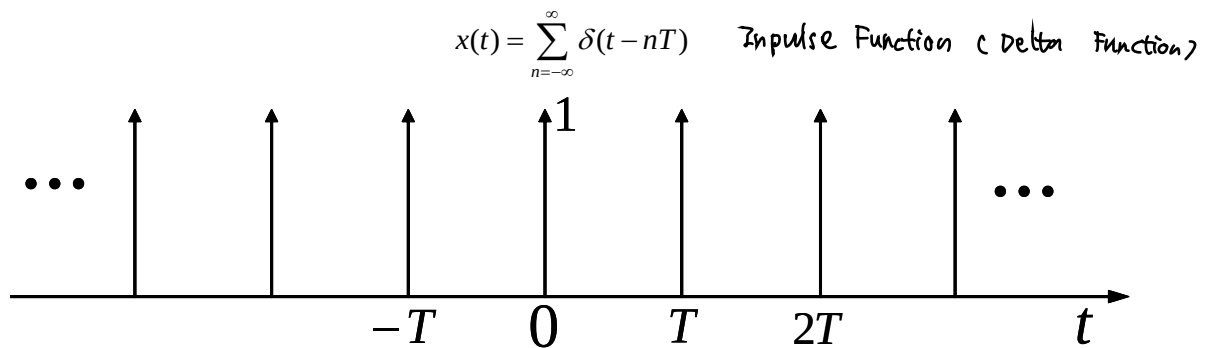
$$C_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-j\omega_0 n t} dt$$

Substituting the values in from our periodic function ($T = 2, \omega_0 = \frac{2\pi}{T} = \pi, f(t) = e^{-t}$) would yield

$$\begin{aligned} C_n &= \frac{1}{2} \int_{-1}^1 e^{-t} e^{-j\omega_0 n t} dt \\ &= \frac{1}{2} \int_0^2 e^{-t} e^{-j\omega_0 n t} dt \\ &= \frac{1}{2} \left[\frac{-1}{1 + j\omega_0 n} \cdot e^{(-1 - j\omega_0 n)t} \right]_0^2 \\ C_n &= \frac{1 - e^{-2(1 + j\omega_0 n)}}{2(1 + j\omega_0 n)} = \frac{1 - e^{-2} e^{-2j\pi n}}{2(1 + j\pi n)} = \frac{1 - e^{-2}}{2(1 + j\pi n)} \end{aligned}$$

Example 2.11

The function $x(t)$ shown in the following figure



is a series of delta functions. It is periodic and hence can be written in a Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}.$$

Determine the fundamental frequency and the Fourier Series coefficients for $x(t)$.

2.7.1. Concept of negative frequency

The exponential Fourier series represents an arbitrary periodic function $x(t)$ by a linear combination of exponential functions of frequencies, $0, \pm\omega_0, \pm2\omega_0, \pm3\omega_0, \dots$. Therefore, there will be negative frequencies in the signals we are studying. However, it should be noted that negative frequencies exist in our formulation only because we want to express $\cos(t)$ and $\sin(t)$ as exponentials. There is no negative frequency signals in real life, rather they are only a

mathematical concept required to give a compact mathematical description of a signal. We will show that it is more convenient to use exponential rather than trigonometric Fourier series representation.

Example 2.12

The periodic square wave $x(t)$ is defined over one period as

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

with period T and is sketched below.

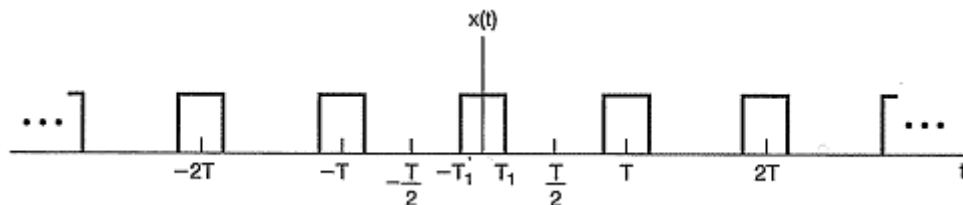


Figure 2.24 Periodic square wave with period T .

Plot the spectrum of

a) $y(t) = x(t)e^{j3\frac{2\pi}{T}t} = x(t)e^{j3\omega_0 t}$;

b) $z(t) = x(t)\cos\left(5\frac{2\pi}{T}t\right) = x(t)\cos(5\omega_0 t)$.

Solution:

a) From Example 2.8, we know that $x(t)$ can be written as a Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

with

$$a_k = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(\omega T_1)}{\omega T} \Big|_{\omega=k\omega_0} = \frac{2T_1}{T} \text{sinc}(2kT_1/T).$$

In this case, $y(t)$ (which is still a periodic function with period T) can be written as

$$\begin{aligned} y(t) &= x(t)e^{j3\omega_0 t} = \left(\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \right) e^{j3\omega_0 t} \\ &= \sum_{k=-\infty}^{\infty} a_k e^{j(k+3)\omega_0 t} \\ &= \sum_{m=-\infty}^{\infty} a_{m-3} e^{jm\omega_0 t} \quad (\text{setting } m = k+3) \end{aligned}$$

If we write $y(t)$ as a Fourier series $y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0 t}$, the Fourier series coefficients are given by $b_k = a_{k-3}$

The spectrum of $y(t)$ is sketched below. Note that the phase of b_k is 0 for all k and is therefore not shown.

Intuitively, we can interpret the results as follows: when a periodic signal is multiplied by a complex exponential, the spectrum will be shifted.

b) Using Euler's formula,

$$\begin{aligned}
z(t) &= x(t) \cos(5\omega_0 t) = x(t) \frac{1}{2} [e^{j5\omega_0 t} + e^{-j5\omega_0 t}] \\
&= \left(\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \right) \frac{1}{2} [e^{j5\omega_0 t} + e^{-j5\omega_0 t}] \\
&= \sum_{k=-\infty}^{\infty} \frac{1}{2} a_k e^{j(k+5)\omega_0 t} + \sum_{k=-\infty}^{\infty} \frac{1}{2} a_k e^{j(k-5)\omega_0 t} \\
&= \sum_{k=-\infty}^{\infty} \frac{1}{2} a_{k-5} e^{jk\omega_0 t} + \sum_{k=-\infty}^{\infty} \frac{1}{2} a_{k+5} e^{jk\omega_0 t} \\
&= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2} (a_{k-5} + a_{k+5}) \right) e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}
\end{aligned}$$

The spectrum of $z(t)$ is sketched below.

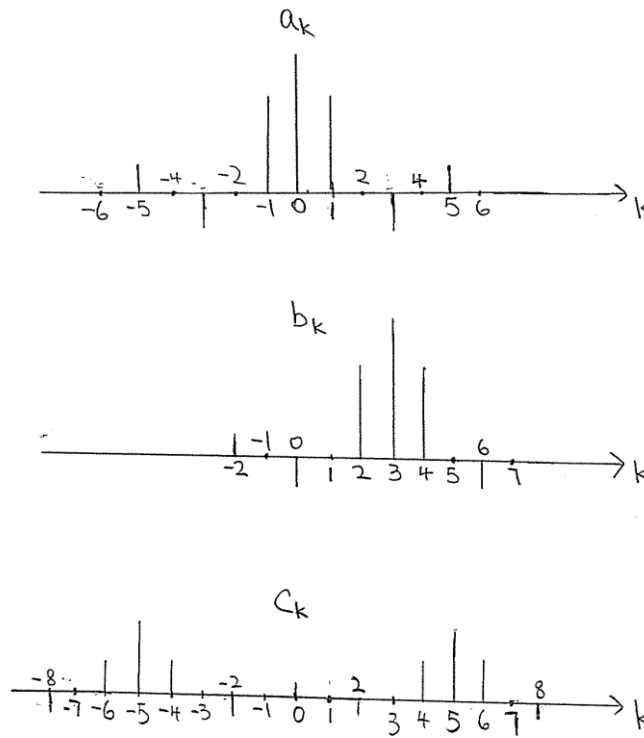


Figure 2.25 Spectrum of $x(t)$, $y(t)$ and $z(t)$ for Example 2.11.

2.7.2. Time domain and frequency domain representation of a signal

A periodic signal $x(t)$ can be specified in two equivalent ways:

1. Time domain representation;

where $x(t)$ is represented as a function of time. Graphical time domain representation is termed as *waveform*.

2. The frequency domain representation;

Where the signal is defined in terms of its *spectrum*. The signals are considered as the superposition of many signals at different frequencies, thus their characteristics can be investigated in the frequency domain.

Any of the above two representations uniquely specifies the function, i.e., if the signal is specified in time domain, we can determine its spectrum. Conversely, if the spectrum is specified, we can determine the corresponding time domain function of a signal.

Signals can be observed in the frequency domain by the use of a Spectrum Analyzer.

2.8 Spectral representation of Aperiodic Signal and the Fourier Transform

We will now consider Fourier series representation for aperiodic signals. We will start by re-examining the continuous-time periodic square wave in Example 2.9. In particular, over one period

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

and periodically repeats with period T , as shown in Figure 2.26.

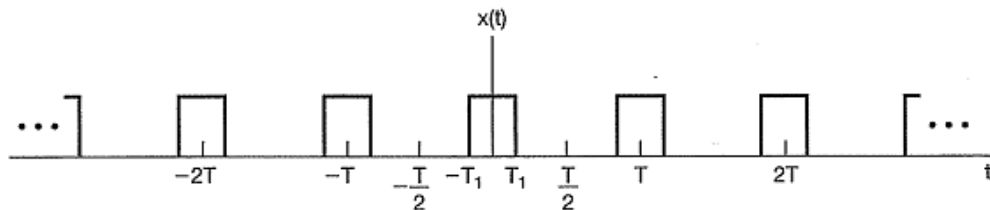


Figure 2.26 A continuous-time periodic square wave.

As determined in Example 2.9, the Fourier series coefficients a_k for this square wave are [from eq. (2.16)]

$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{2T_1}{T} \text{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right), \quad (2.19)$$

where $\omega_0 = 2\pi/T$. In Figure 2.27, bar graphs of these coefficients were shown for a fixed value of $T_1 = T/4$.

An alternative way of interpreting eq.(2.19) is as samples of a sinc function. Specifically, we can write

Effect of changing period
Discrete $\rightarrow$ continuous

$$Ta_k = 2T_1 \text{sinc}\left(\frac{\omega T_1}{\pi}\right) \bigg|_{\omega=k\omega_0} = \frac{2 \sin \omega T_1}{\omega} \bigg|_{\omega=k\omega_0} \quad (2.20)$$

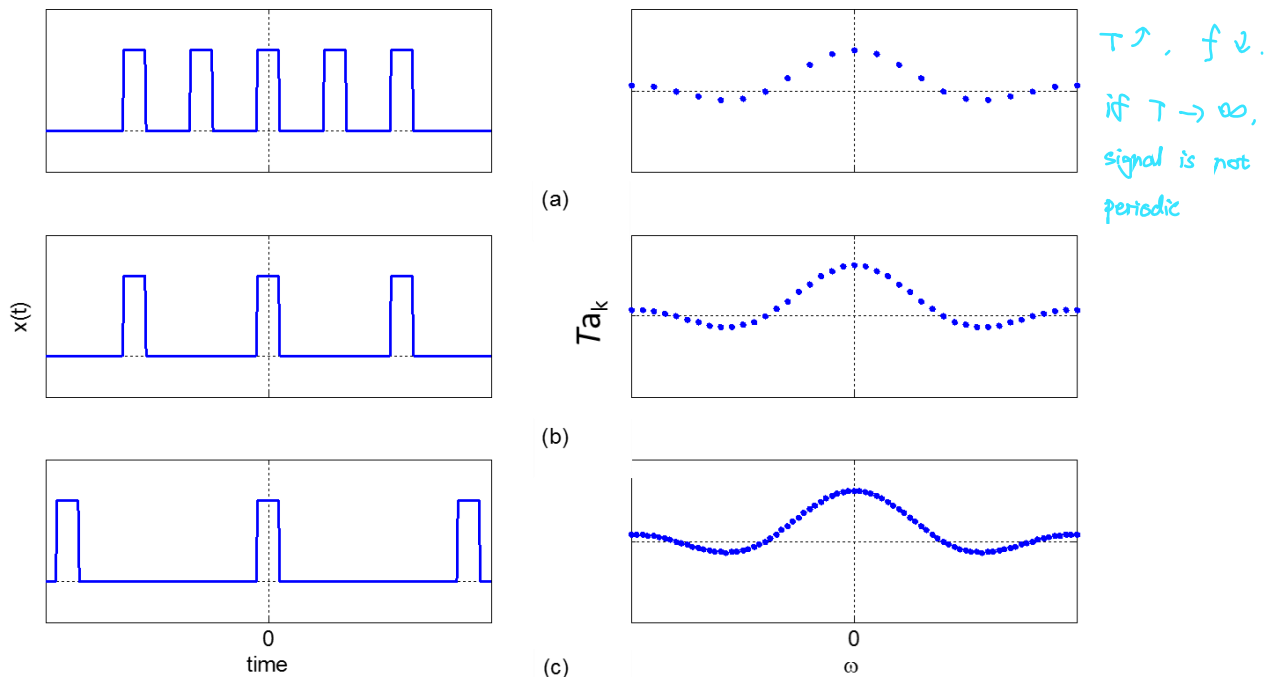


Figure 2.27 The Fourier series coefficients and their envelope for the periodic square wave in Figure 2.26 for several values of T (with T_1 fixed): (a) $T = 8T_1$; (b) $T = 16T_1$; (c) $T = 32T_1$.

The function $(2 \sin \omega T_1) / \omega$ can be thought of as the envelope of Ta_k , and the coefficients a_k as evenly spaced samples of the envelope, if ω is thought of as being a continuous variable. In Figure 2.27, the relationship between the periodicity of $x(t)$ and its corresponding envelope (eq.(2.20)) is shown. It is noticeable that as the periodicity T

increases (or equivalently, as the fundamental frequency $\omega_0 = 2\pi/T$ decreases), the periodic square wave approaches a single rectangular pulse, an aperiodic signal of just 1 period, rather than infinite. It is also noticeable how the Fourier series coefficients are more closely packed together as the periodicity T increases. From this trend, we notice that the Fourier series coefficients approach the continuous envelope function as $T \rightarrow \infty$.

The aforementioned example describes Fourier's development at describing aperiodic signals. We can think of an aperiodic signal as a periodic signal whose periodicity grows to infinity, and observe this limit on the Fourier series description based on this signal. Let us observe an arbitrary aperiodic signal $x(t)$ of a finite duration. That is, for some number T_1 , $x(t) = 0$ if $|t| \geq T_1$, as illustrated in **Figure 2.28(a)**. From this aperiodic signal, we can construct a periodic signal $\tilde{x}(t)$ for which $x(t)$ is one period, as indicated in **Figure 2.28 (b)**. As we choose the period T to be larger, $\tilde{x}(t)$ is identical to $x(t)$ over a longer interval, and as $T \rightarrow \infty$, $\tilde{x}(t)$ is equal to $x(t)$ for any finite value of t .

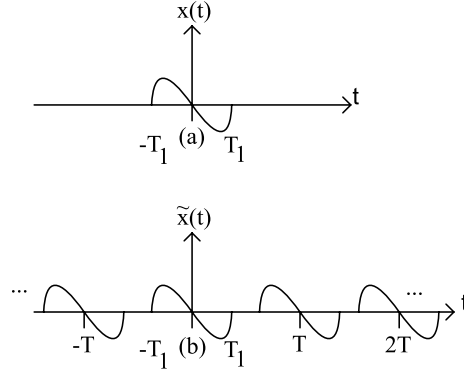


Figure 2.28 (a) Aperiodic signal $x(t)$; (b) periodic signal $\tilde{x}(t)$, constructed to be equal to $x(t)$ over one period.

Let us now examine the effect of this on the Fourier series representation of $\tilde{x}(t)$. For $-T/2 \leq t \leq T/2$, we have

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}, \quad (2.21)$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt, \quad (2.22)$$

where $\omega_0 = 2\pi/T$. Since $\tilde{x}(t) = x(t)$ for $|t| < T/2$, and also, since $x(t) = 0$ outside of this interval, eq. (2.22) can be rewritten as

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt.$$

Therefore, defining the envelope $X(\omega)$ of Ta_k as

$$X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt, \quad (2.23)$$

we have, for the coefficients a_k ,

$$a_k = \frac{1}{T} X(k\omega_0). \quad (2.24)$$

Combining eqs.(2.24) and (2.21), we can express $\tilde{x}(t)$ in terms of $X(\omega)$ as

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{T} X(k\omega_0) e^{jk\omega_0 t},$$

or equivalently, since $2\pi/T = \omega_0$,

$$\tilde{x}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(k\omega_0) e^{jk\omega_0 t} \omega_0. \quad (2.25)$$

As $T \rightarrow \infty$, $\tilde{x}(t)$ approaches $x(t)$, and consequently, in the limit eq. (2.25) becomes a representation of $x(t)$.

Furthermore, $\omega_0 \rightarrow 0$ as $T \rightarrow \infty$, and the right-hand side of eq. (2.25) passes to an integral. This can be seen by considering the graphical interpretation of the equation, illustrated in **Figure 2.29**. Each term in the summation on the right-hand side is the area of a rectangle of height $X(k\omega_0)e^{jk\omega_0 t}$ and width ω_0 (Here, t is regarded as fixed.) As $\omega_0 \rightarrow 0$, the summation converges to the integral of $X(\omega)e^{j\omega t}$. Therefore, using the fact that $\tilde{x}(t) \rightarrow x(t)$ as $T \rightarrow \infty$, we see that eqs. (2.25) and (2.23) respectively become

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) e^{j\omega t} d\omega \quad (2.26)$$

and

$$X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt. \quad (2.27)$$

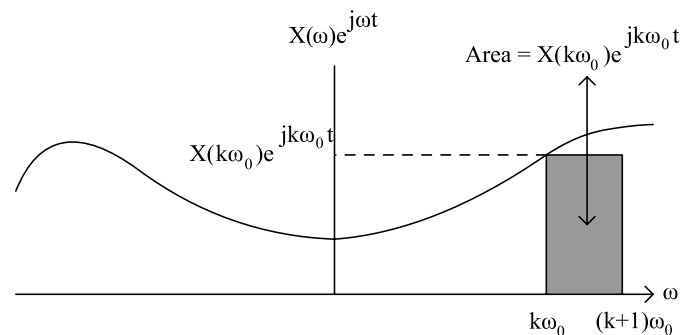


Figure 2.29. Graphical interpretation of eq. (2.25)

Equations (2.26) and (2.27) are referred to as the *Fourier transform pair*; with the function $X(\omega)$ referred to as the *Fourier Transform* or *Fourier integral* of $x(t)$ and eq.(2.26) as the *inverse Fourier transform* equation. The *synthesis* eq.(2.27) plays a role for aperiodic signals similar to that of periodic signals, since both represent a signal as a linear combination of complex exponentials. For periodic signals, these complex exponentials have amplitudes $\{a_k\}$, as given by(2.14), and occur at a discrete set of harmonically related frequencies $k\omega_0$, $k = 0, \pm 1, \pm 2, \dots$. For aperiodic signals, the complex exponentials occur at a continuum of frequencies and, according to the synthesis equation(2.26), have "amplitude" $X(\omega)(d\omega / 2\pi)$. In analogy with the terminology used for the Fourier series coefficients of a periodic signal, the transform $X(\omega)$ of an aperiodic signal $x(t)$ is commonly referred to as the spectrum of $x(t)$, as it provides us with the information needed for describing $x(t)$ as a linear combination (specifically, an integral) of sinusoidal signals at different frequencies.

The following is a way to interpret the concept of Fourier Transform described in (2.26).

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) e^{j\omega t} d\omega$$

1) Any signal

2) is

3) a sum of

4) a lot of exponentials (sinusoids) with different frequency

5) scaled by different coefficients

If you read the interpretations of the above equation from 1) to 5), it says "Any signal is a sum of a lot of exponentials (sinusoids) with different frequencies scaled by different coefficients.

Based on the above development, or equivalently on a comparison of eq.(2.27) and (2.14), we also note that the Fourier coefficients a_k of a periodic signal $\tilde{x}(t)$ can be expressed in terms of equally spaced *samples* of the Fourier transform of one period of $\tilde{x}(t)$. Specifically, suppose that $\tilde{x}(t)$ is a periodic signal with period T and Fourier coefficients a_k . Let $x(t)$ be a finite-duration signal that is equal to $\tilde{x}(t)$ over exactly one period—say, for $s \leq t \leq s+T$ for some values of s —and that is zero otherwise. Then, since (2.22) allows us to compute the Fourier coefficients of $\tilde{x}(t)$ by integrating over any period, we can write

$$a_k = \frac{1}{T} \int_s^{s+T} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_s^{s+T} x(t) e^{-jk\omega_0 t} dt.$$

Since $x(t)$ is zero outside the range $s \leq t \leq s+T$ we can equivalently write

$$a_k = \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt.$$

Comparing with eq.(2.27) we conclude that

$$a_k = \frac{1}{T} X(\omega) \Big|_{\omega=k\omega_0}, \quad (2.28)$$

where $X(\omega)$ is the Fourier transform of $x(t)$. eq.(2.28) states that the Fourier coefficients of $\tilde{x}(t)$ are proportional to samples of the Fourier transform of one period of $\tilde{x}(t)$.

Example 2.13

Consider the signal

$$x(t) = e^{-bt} u(t) \quad b > 0.$$

From eq. (2.27),

$$X(\omega) = \int_0^{\infty} e^{-bt} e^{-j\omega t} dt = -\frac{1}{b+j\omega} e^{-(b+j\omega)t} \Big|_0^{\infty}$$

That is,

$$X(\omega) = \frac{1}{b+j\omega}, \quad b > 0.$$

Since this Fourier transform is complex valued, to plot it as a function of ω , we express $X(\omega)$ in terms of its magnitude and phase:

$$|X(\omega)| = \frac{1}{\sqrt{b^2 + \omega^2}}, \quad \angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{b}\right).$$

Each of these components is sketched in **Figure 2.30**. Note that if b is complex rather than real, then $x(t)$ is absolutely integrable as long as $\Re\{b\} > 0$ and in this case the preceding calculation yields the same form for $X(\omega)$.

That is,

$$X(\omega) = \frac{1}{b+j\omega}, \quad \Re\{b\} > 0.$$

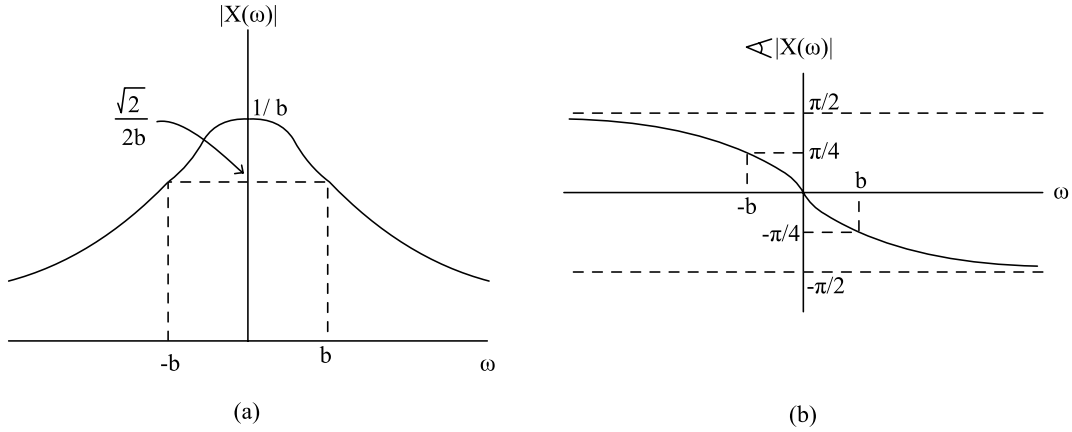


Figure 2.30 Fourier transform of the signal $x(t) = e^{-bt}u(t)$, $b > 0$, considered in Example 2.13.

Example 2.14

Let

$$x(t) = e^{-b|t|}, \quad b > 0.$$

This signal is sketched in **Figure 2.31(a)**. The Fourier transform of the signal is

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{+\infty} e^{-b|t|} e^{-j\omega t} dt \\ &= \int_{-\infty}^0 e^{bt} e^{-j\omega t} dt + \int_0^{+\infty} e^{-bt} e^{-j\omega t} dt \\ &= \frac{1}{b - j\omega} + \frac{1}{b + j\omega} \\ &= \frac{2b}{b^2 + \omega^2}. \end{aligned}$$

In this case $X(\omega)$ is real, and it is illustrated in **Figure 2.31(b)**.

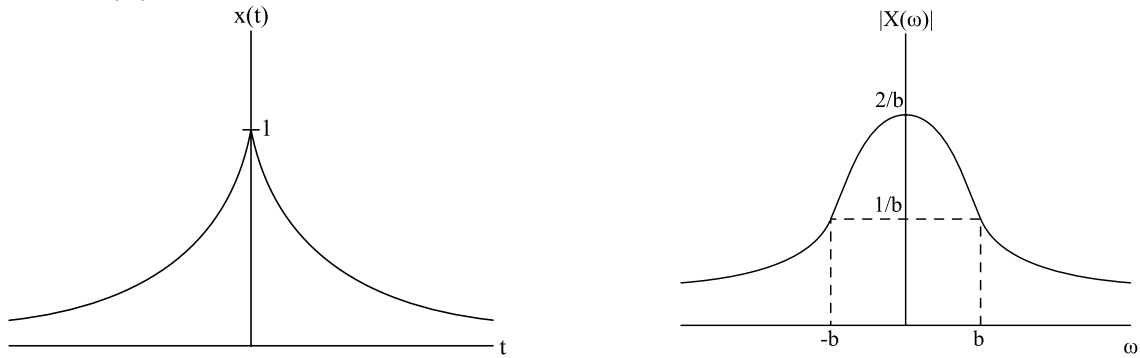
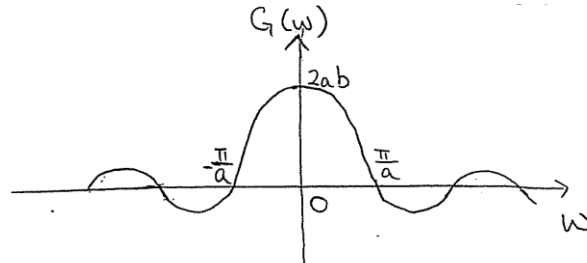
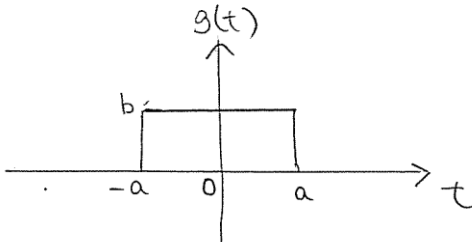


Figure 2.31. (a) Signal $x(t) = e^{-b|t|}$ of Example 2.14 and (b) its Fourier Transform $X(\omega)$.

2.9 Fourier transform of some useful functions

2.9.1. Rectangular function:

$$g(t) = \begin{cases} b, & |t| < a \\ 0, & |t| > a \end{cases}$$

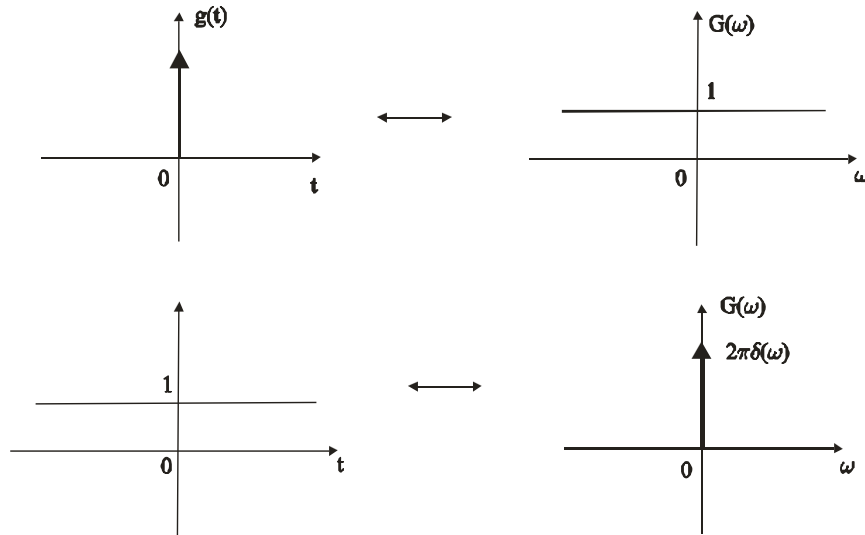


The Fourier Transform of $g(t)$ is given by

$$G(\omega) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t} dt = \int_{-a}^a be^{-j\omega t} dt = b \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-a}^a = b \frac{e^{-j\omega a} - e^{j\omega a}}{-j\omega} = b \frac{2j \sin(\omega a)}{j\omega} = \frac{2b}{\omega} \sin(\omega a) = 2ab \operatorname{sinc}\left(\frac{\omega a}{\pi}\right)$$

2.9.2. Unit impulse function:

$$\delta(t) \Leftrightarrow 1 \quad \text{and} \quad 1 \Leftrightarrow 2\pi\delta(\omega)$$



Proof:

$$F[\delta(t)] = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t)e^0 dt = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$F^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = \frac{1}{2\pi}$$